

# Module 03 Assignments

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## Group 7.6

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### Question 1

$$\int_0^{\frac{\pi}{2}} \cos(5x) \cos(2x) dx; \text{ entry 80}$$

Entry 80

$$\int \cos(au) \cos(bu) du = \frac{\sin(a-b)u}{2(a-b)} + \frac{\sin(a+b)u}{2(a+b)} + C$$

Identify  $a$  and  $b$

$$a = 5 \quad b = 2$$

Substitute

$$\int \cos(5x) \cos(2x) dx = \frac{\sin(3x)}{6} + \frac{\sin(7x)}{14} + C$$

Evaluate at the upper bound

$$\frac{\sin(3 \cdot \frac{\pi}{2})}{6} + \frac{\sin(7 \cdot \frac{\pi}{2})}{14}$$

$$= \frac{\sin(\frac{3\pi}{2})}{6} + \frac{\sin(\frac{7\pi}{2})}{14}$$

$$-\frac{1}{6} - \frac{1}{14}$$

$$-\frac{7}{42} - \frac{3}{42}$$

$$-\frac{5}{21}$$

Evaluate at the lower bound

$$= \frac{\sin(0)}{6} + \frac{\sin(0)}{14} = 0$$

Subtract

$$-\frac{5}{21} - 0$$

$$\boxed{-\frac{5}{21}}$$


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## Question 2

$$\int_0^1 \sqrt{x-x^2} \, dx; \text{ entry 113}$$

Entry 113

$$\sqrt{2au-u^2} \, du = \frac{u-a}{2} \sqrt{2au-u^2} + \frac{a^2}{2} \arccos\left(\frac{a-u}{a}\right) + C$$

Identify  $u$  and  $a$

$$u = x$$

Compare

$$x-x^2 \qquad 2ax-x^2$$

So

$$2a = 1$$

therefore

$$a = \frac{1}{2}$$

Substitute into the formula

$$\sqrt{2\left(\frac{1}{2}\right)x - x^2} dx = \frac{x - \frac{1}{2}}{2} \sqrt{2\left(\frac{1}{2}\right)x - x^2} + \frac{\left(\frac{1}{2}\right)^2}{2} \arccos\left(\frac{\frac{1}{2} - x}{\frac{1}{2}}\right) + C$$

Evaluate at the upper bound  $x = 1$

$$\frac{1 - \frac{1}{2}}{2} \sqrt{2\left(\frac{1}{2}\right)1 - 1} + \frac{\left(\frac{1}{2}\right)^2}{2} \arccos\left(\frac{\frac{1}{2} - 1}{\frac{1}{2}}\right)$$

$$= 0 + \frac{1}{8} \arccos(-1)$$

$$= 0 + \frac{\pi}{8}$$

$$\frac{\pi}{8}$$

Evaluate at the lower bound  $x = 0$

$$\frac{0 - \frac{1}{2}}{2} \sqrt{0} + \frac{\left(\frac{1}{2}\right)^2}{2} \arccos\left(\frac{\frac{1}{2} - 0}{\frac{1}{2}}\right)$$

$$= 0 + \frac{1}{8} \arccos(1)$$

$$= 0$$

Subtract

$$\frac{\pi}{8} - 0$$

$$\boxed{\frac{\pi}{8}}$$

## Question 6

$$\int \frac{\sqrt{t^6 - 5}}{t} dt; \text{ entry 41}$$

Entry 41

$$\int \frac{\sqrt{u^2 - a^2}}{u} du = \sqrt{u^2 - a^2} - a \arccos\left(\frac{a}{|u|}\right) + C$$

Choose  $u$

Since

$$t^6 = (t^3)^2$$

let

$$u = t^3$$

Then

$$du = 3t \, dt$$

and

$$dt = \frac{du}{3t}$$

Rewrite

$$\frac{dt}{t} = \frac{1}{t} \left( \frac{du}{3t} \right) = \frac{du}{3t^3}$$

Since

$$u = t^3$$

we have

$$\frac{dt}{t} = \frac{du}{3u}$$

Rewrite the integral

$$\int \frac{\sqrt{t^6 - 5}}{t} dt = \frac{1}{3} \int \frac{\sqrt{u^2 - 5}}{u} du$$

Identify  $a$

$$a = \sqrt{5}$$

Apply entry 41

$$\frac{1}{3} \left[ \sqrt{u^2 - 5} - \sqrt{5} \arccos \left( \frac{\sqrt{5}}{|u|} \right) \right] + C$$

Distribute  $\frac{1}{3}$

$$\frac{1}{3}\sqrt{u^2-5}-\frac{\sqrt{5}}{3}\arccos\left(\frac{\sqrt{5}}{|u|}\right)+C$$

Substitute  $u = t^3$

$$\frac{1}{3}\sqrt{t^6-5}-\frac{\sqrt{5}}{3}\arccos\left(\frac{\sqrt{5}}{|t^3|}\right)+C$$

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### Question 8

$$\int_0^2 x^2\sqrt{4-x^2} \, dx$$

Entry 31

$$\int u^2\sqrt{a^2-u^2} \, du = \frac{u}{8}(2u^2-a^2)\sqrt{a^2-u^2} + \frac{a^4}{8}\arcsin\left(\frac{u}{a}\right) + C$$

Identify  $u$  and  $a$

$$u = x \quad a = 2$$

Substitute into the formula

$$\int x^2\sqrt{4-x^2} \, dx = \frac{x}{8}(2x^2-2^2)\sqrt{2^2-x^2} + \frac{2^4}{8}\arcsin\left(\frac{x}{2}\right) + C$$

Simplify

$$\frac{x}{8}(2x^2-4)\sqrt{4-x^2} + 2\arcsin\left(\frac{x}{2}\right)$$

Evaluate at the upper bound  $x = 2$

$$\frac{x}{8}(2(2)^2-4)\sqrt{4-2^2} + 2\arcsin\left(\frac{2}{2}\right)$$

$$= 0 + 2\arcsin(1)$$

$$= \pi$$

Evaluate at the lower bound  $x = 0$

$$\frac{0}{8} (2(0)^2 - 4) \sqrt{4 - 0^2} + 2 \arcsin \left( \frac{0}{2} \right)$$

$$= 0 + 2 \arcsin (0)$$

$$= 0$$

Subtract

$$\pi - 0$$

$$\boxed{\pi}$$


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### Question 10

$$\int \frac{e^x}{4 - e^{2x}} dx$$

Choose  $u$

Since

$$e^{2x} = (e^x)^2$$

let

$$u = e^x$$

Then

$$du = e^x dx$$

So the integral becomes

$$\int \frac{du}{4 - u^2}$$

Use entry 19

$$\int \frac{du}{a^2 - u^2} = \frac{1}{2a} \ln \left| \frac{u + a}{u - a} \right| + C$$

Substitute  $a = 2$

$$\int \frac{du}{4-u^2} = \frac{1}{4} \ln \left| \frac{u+2}{u-2} \right| + C$$

Substitute  $u = e^x$  back in

$$\boxed{\frac{1}{4} \ln \left| \frac{e^x + 2}{e^x - 2} \right| + C}$$


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### Question 11

$$\int \frac{\sqrt{9x^2 + 4}}{x^2} dx$$

Choose  $u$

Since

$$9x^2 = (3x)^2$$

let

$$u = 3x \quad du = 3 dx \quad \frac{du}{3} = dx$$

So

$$x = \frac{u}{3} \quad x^2 = \frac{u^2}{9}$$

Rewrite the integral

$$\begin{aligned} & \int \frac{\sqrt{u^2 + 4}}{\frac{u^2}{9}} \left( \frac{du}{3} \right) \\ &= 3 \int \frac{\sqrt{u^2 + 4}}{u^2} du \end{aligned}$$

Match the table entry to entry 24

$$\int \frac{\sqrt{a^2 + u^2}}{u} du = -\frac{\sqrt{a^2 + u^2}}{u} + \ln \left( u + \sqrt{a^2 + u^2} \right) + C$$

Since

$$a^2 = 4 \quad a = 2$$

Apply entry 24

$$3 \left[ -\frac{\sqrt{u^2 + 4}}{u^2} + \ln(u + \sqrt{u^2 + 4}) \right] + C$$

Distribute the 3

$$-\frac{3\sqrt{u^2 + 4}}{u} + 3 \ln(u + \sqrt{u^2 + 4}) + C$$

Substitute  $u = 3x$  back in

$$-\frac{3\sqrt{(3x)^2 + 4}}{3x} + 3 \ln(3x + \sqrt{(3x)^2 + 4}) + C$$

Simplify

$$-\frac{\sqrt{9x^2 + 4}}{x} + 3 \ln(3x + \sqrt{9x^2 + 4}) + C$$

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### Question 13

$$\int_0^\pi \cos^6(\theta) d\theta$$

Use entry 74

$$\int \cos^n(u) du = \frac{1}{n} \cos^{n-1}(u) \sin(u) + \frac{n-1}{n} \int \cos^{n-2}(u) du$$

For  $n = 6$

$$\int \cos^6(\theta) d\theta = \frac{1}{6} \cos^5(\theta) \sin(\theta) + \frac{5}{6} \int \cos^4(\theta) d\theta$$

Apply entry 74 again

$$\int \cos^4(\theta)$$

For  $n = 4$

$$\int \cos^4(\theta) d\theta = \frac{1}{4} \cos^3(\theta) \sin(\theta) + \frac{3}{4} \int \cos^2(\theta) d\theta$$

Substitute

$$\int \cos^6(\theta) d\theta = \frac{1}{6} \cos^5(\theta) \sin(\theta) + \frac{5}{6} \left[ \frac{1}{4} \cos^3(\theta) \sin(\theta) + \frac{3}{4} \int \cos^2(\theta) d\theta \right]$$

Distribute

$$= \frac{1}{6} \cos^5(\theta) \sin(\theta) + \frac{5}{24} \cos^3(\theta) \sin(\theta) + \frac{5}{8} \int \cos^2(\theta) d\theta$$

Apply the half-angle identity

$$\cos^2(\theta) = \frac{1 + \cos(2\theta)}{2}$$

So

$$\int \cos^2(\theta) d\theta = \frac{1}{2} \theta + \frac{1}{4} \sin(2\theta)$$

Substitute

$$\frac{1}{6} \cos^5(\theta) \sin(\theta) + \frac{5}{24} \cos^3(\theta) \sin(\theta) + \frac{5}{8} \left( \frac{1}{2} \theta + \frac{1}{4} \sin(2\theta) \right)$$

Simplify

$$\frac{1}{6} \cos^5(\theta) \sin(\theta) + \frac{5}{24} \cos^3(\theta) \sin(\theta) + \frac{5}{16} \theta + \frac{5}{32} \sin(2\theta)$$

Evaluate at the upper bound

$$\theta = \pi$$

$$\frac{1}{6}(-1)^5(0) + \frac{5}{24}(-1)^3(0) + \frac{5\pi}{16} + \frac{5}{32}(0)$$

$$= \frac{5\pi}{16}$$

Evaluate at the lower bound

$$\theta = 0$$

$$\frac{1}{6}(1)^5(0) + \frac{5}{24}(1)^3(0) + \frac{5}{16}(0) + \frac{5}{32}(0)$$

$$= 0$$

Subtract

$$\frac{5\pi}{16} - 0$$

$$\boxed{\frac{5\pi}{16}}$$

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### Question 15

$$\int \frac{\arctan(\sqrt{x})}{\sqrt{x}} dx$$

Choose  $u$

Let

$$u = \sqrt{x}$$

Differentiate

$$dx = 2u \, du$$

Rewrite the differential factor

$$\int \frac{dx}{\sqrt{x}} = \frac{2u \, du}{u} = 2 \, du$$

The integral becomes

$$2 \int \arctan(u) \, du$$

Use entry 95

$$\int u^n \arctan(u) \, du = \frac{1}{n+1} \left[ u^{n+1} \arctan(u) - \int \frac{u^{n+1} \, du}{1+u^2} \right]$$

Because  $u^0 = 1$ ,  $n = 0$

$$2 \int \arctan(u) \, du = 2 \left[ u \arctan(u) - \int \frac{u}{1+u^2} \, du \right]$$

Distribute

$$2u \arctan(u) - 2 \int \frac{u}{1+u^2} du$$

Evaluate the remaining integral

Let

$$w = 1 + u^2 \quad dw = 2u \, du \quad u \, du = \frac{1}{2} dw$$

Therefore

$$\begin{aligned} 2 \int \frac{u}{1+u^2} du &= 2 \left( \frac{1}{2} \right) \int \frac{1}{w} dw \\ &= \ln(1+u^2) \end{aligned}$$

So

$$2 \int \arctan(u) \, du = 2u \arctan(u) - \ln(1+u^2) + C$$

Substitute  $u = \sqrt{x}$  back in

$$\boxed{2\sqrt{x} \arctan(\sqrt{x}) - \ln(1+x) + C}$$


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### Question 23

$$\int \frac{\sin(2\theta)}{\sqrt{\cos^4(\theta) + 4}} d\theta$$

Choose  $u$

Let

$$u = \cos^2(\theta) \quad du = -2 \sin(\theta) \cos(\theta) \, d\theta \quad \sin(2\theta) = 2 \sin(\theta) \cos(\theta)$$

Therefore

$$\sin(2\theta) \, d\theta = -du$$

Rewrite the integral

$$- \int \frac{du}{\sqrt{u^2 + 4}}$$

Apply entry 25

$$\int \frac{du}{\sqrt{a^2 + u^2}} = \ln \left( u + \sqrt{a^2 + u^2} \right) + C$$
$$- \left( \ln \left( u + \sqrt{u^2 + 4} \right) \right) + C$$

Substitute  $u = \cos^2(\theta)$  back in

$$- \ln \left( \cos^2(\theta) + \sqrt{(\cos^2(\theta))^2 + 4} \right) + C$$

Simplify

$$\boxed{- \ln \left( \cos^2(\theta) + \sqrt{\cos^4(\theta) + 4} \right) + C}$$

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### Question 25

$$\int x^3 e^{2x} dx$$

Apply entry 97

$$\int u^n e^{au} du = \frac{1}{a} u^n e^{au} - \frac{n}{a} \int u^{n-1} e^{au} du$$

$$u = x \quad n = 3 \quad a = 2$$

$$\frac{1}{2} x^3 e^{2x} - \frac{3}{2} \int x^2 e^{2x} dx$$

Apply entry 97 again

$$\int x^2 e^{2x} dx$$

$$n = 2 \quad a = 2$$

$$\frac{1}{2} x^2 e^{2x} - \int x e^{2x} dx$$

Apply entry 97 again

$$\int x e^{2x} dx$$

$$n = 1 \quad a = 2$$

$$\frac{1}{2} x e^{2x} - \frac{1}{2} \int e^{2x} dx$$

Evaluate the integral

$$\int e^{2x} dx = \frac{1}{2} e^{2x}$$

we get

$$\int x e^{2x} dx = \frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x}$$

Substitute back

$$\int x^2 e^{2x} dx = \frac{1}{2} x^2 e^{2x} - \left( \frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} \right)$$

Distribute

$$\int x^2 e^{2x} dx = \frac{1}{2} x^2 e^{2x} - \frac{1}{2} x e^{2x} + \frac{1}{4} e^{2x}$$

Substitute this into the original reduction

$$\int x^3 e^{2x} dx = \frac{1}{2} x^3 e^{2x} - \frac{3}{2} \left( \frac{1}{2} x^2 e^{2x} - \frac{1}{2} x e^{2x} + \frac{1}{4} e^{2x} \right)$$

Distribute

$$= \frac{1}{2} x^3 e^{2x} - \frac{3}{4} x^2 e^{2x} + \frac{3}{4} x e^{2x} - \frac{3}{8} e^{2x}$$

Factor out  $\frac{1}{8} e^{2x}$

$$\frac{1}{8} e^{2x} (4x^3 - 6x^2 + 6x - 3) + C$$

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### Question 26

$$\int x^3 \arcsin(x^2) dx$$

Apply  $u$ -substitution

$$u = x^2 \quad du = 2x dx$$

Rewrite  $x^3 dx$

$$x^3 dx = x^2 x dx$$

$$x^3 dx = u \left( \frac{1}{2} du \right)$$

Therefore

$$\frac{1}{2} \int u \arcsin(u) du$$

Apply entry 93 with  $n = 1$

$$\frac{1}{4} \left[ u^2 \arcsin(u) - \int \frac{u^2}{\sqrt{1-u^2}} du \right]$$

Rewrite the remaining integral

Since

$$u^2 = 1 - (1 - u^2)$$

$$\int \frac{u^2}{\sqrt{1-u^2}} du = \int \frac{1}{\sqrt{1-u^2}} du - \int \sqrt{1-u^2} du$$

Apply the inverse-sine form and entry 30

$$= \arcsin(u) - \left( \frac{u}{2} \arcsin(u) - \frac{u}{2} \sqrt{1-u^2} \right)$$

$$= \frac{1}{2} \arcsin(u) - \frac{u}{2} \sqrt{1-u^2}$$

Substitute into the original expression

$$\frac{1}{4} \left[ u^2 \arcsin(u) - \left( \frac{1}{2} \arcsin(u) - \frac{u}{2} \sqrt{1-u^2} \right) \right] + C$$

Simplify

$$\frac{u^2}{4} \arcsin(u) - \frac{1}{8} \arcsin(u) + \frac{u}{8} \sqrt{1-u^2} + C$$

Substitute  $u = x^2$

$$\boxed{\frac{x^4}{4} \arcsin(x^2) - \frac{1}{8} \arcsin(x^2) + \frac{x^2}{8} \sqrt{1-x^4} + C}$$

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## Group 7.7

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### Question 7

$$\int_0^1 \sqrt{1+x^3} dx, n=4$$

Include the intermediate steps that lead to the answer.

Find  $\Delta x$

$$\Delta x = \frac{b-a}{n} = \frac{1-0}{4} = \frac{1}{4}$$

Therefore

$$x_0 = 0, \quad x_1 = \frac{1}{4}, \quad x_2 = \frac{1}{2}, \quad x_3 = \frac{3}{4}, \quad x_4 = 1$$

The function values are

$$f(0) = 1$$

$$f\left(\frac{1}{4}\right) = \sqrt{\frac{65}{64}} = 1.007782219$$

$$f\left(\frac{1}{2}\right) = \sqrt{\frac{9}{8}} = 1.060660172$$

$$f\left(\frac{3}{4}\right) = \sqrt{\frac{91}{64}} = 1.192424002$$

$$f(1) = \sqrt{2} = 1.414213562$$

**a.** the Trapezoidal Rule

$$T_4 = \frac{\Delta x}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)]$$

Substitute

$$T_4 = \frac{1}{8} [1 + 2(1.007782219) + 2(1.060660172) + 2(1.192424002) + 1.414213562]$$

Therefore

$$T_4 \approx 1.116993$$

**b.** the Midpoint Rule

The midpoints are

$$\frac{1}{8}, \quad \frac{3}{8}, \quad \frac{5}{8}, \quad \frac{7}{8}$$

Their function values are

$$f\left(\frac{1}{8}\right) = \sqrt{\frac{513}{512}} = 1.000976086$$

$$f\left(\frac{3}{8}\right) = \sqrt{\frac{539}{512}} = 1.026028447$$

$$f\left(\frac{5}{8}\right) = \sqrt{\frac{637}{512}} = 1.115410519$$

$$f\left(\frac{7}{8}\right) = \sqrt{\frac{855}{512}} = 1.292254571$$

Substitute the calculated values in the midpoint rule.

$$M_4 = \frac{1}{4} [1.000976086 + 1.026028447 + 1.115410519 + 1.292254571]$$

Therefore

$$M_4 \approx 1.108667$$

**c.** Simpson's Rule to approximate the given integral with the specified value of  $n$ .

The coefficient pattern for  $n = 4$  is

$$1, 4, 2, 4, 1$$

So

$$S_4 = \frac{\Delta x}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)]$$

Since

$$\frac{\Delta x}{3} = \frac{\frac{1}{4}}{3} = \frac{1}{12}$$

We have

$$S_4 = \frac{1}{12} [1 + 4(1.007782219) + 2(1.060660172) + 4(1.192424002) + 1.414213562]$$

Therefore

$S_4 \approx 1.111363$

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### Question 9

$$\int_0^1 \sqrt{e^x - 1} \, dx, \, n = 10$$

Include the intermediate steps that lead to the answer.

Find  $\Delta x$

$$\Delta x = \frac{b - a}{n} = \frac{1 - 0}{10} = \frac{1}{10}$$

The  $x_i$  values are

$$x_0 = 0, \quad x_1 = \frac{1}{10}, \quad x_2 = \frac{2}{10}, \quad x_3 = \frac{3}{10}, \quad x_4 = \frac{4}{10}$$

$$x_5 = \frac{5}{10}, \quad x_6 = \frac{6}{10}, \quad x_7 = \frac{7}{10}, \quad x_8 = \frac{8}{10}, \quad x_9 = \frac{9}{10}$$

$$x_{10} = 1$$

The function values are

$$f(0) = \sqrt{e^0 - 1} = 0$$

$$f\left(\frac{1}{10}\right) = \sqrt{e^{\frac{1}{10}} - 1} = 0.32430066$$

$$f\left(\frac{2}{10}\right) = \sqrt{e^{\frac{2}{10}} - 1} = 0.4705345451$$

$$f\left(\frac{3}{10}\right) = \sqrt{e^{\frac{3}{10}} - 1} = 0.5914886369$$

$$f\left(\frac{4}{10}\right) = \sqrt{e^{\frac{4}{10}} - 1} = 0.7013021443$$

$$f\left(\frac{5}{10}\right) = \sqrt{e^{\frac{5}{10}} - 1} = 0.8054323502$$

$$f\left(\frac{6}{10}\right) = \sqrt{e^{\frac{6}{10}} - 1} = 0.9067076709$$

$$f\left(\frac{7}{10}\right) = \sqrt{e^{\frac{7}{10}} - 1} = 1.006852873$$

$$f\left(\frac{8}{10}\right) = \sqrt{e^{\frac{8}{10}} - 1} = 1.107041521$$

$$f\left(\frac{9}{10}\right) = \sqrt{e^{\frac{9}{10}} - 1} = 1.208140352$$

$$f(1) = \sqrt{e^1 - 1} = 1.310832494$$

**a.** the Trapezoidal Rule

$$T_{10} = \frac{\Delta x}{2} [f(x_0) + 2f(x_1) + \cdots + 2f(x_9) + f(x_{10})]$$

Substitute

$$\begin{aligned} T_{10} = \frac{1}{20} & \left[ 0 + 2(0.32430066) + 2(0.4705345451) + 2(0.5914886369) + 2(0.7013021443) \right. \\ & + 2(0.8054323502) + 2(0.9067076709) + 2(1.006852873) \\ & \left. + 2(1.107041521) + 2(1.208140352) + 1.310832494 \right] \end{aligned}$$

Therefore

$$\boxed{T_{10} \approx 0.777722}$$

**b.** the Midpoint Rule

The midpoints are

$$\frac{1}{20}, \quad \frac{3}{20}, \quad \frac{5}{20}, \quad \frac{7}{20}, \quad \frac{9}{20}, \quad \frac{11}{20}, \quad \frac{13}{20}, \quad \frac{15}{20}, \quad \frac{17}{20}, \quad \frac{19}{20}$$

Their function values are

$$f(0) = \sqrt{e^0 - 1} = 0$$

$$f\left(\frac{1}{20}\right) = \sqrt{e^{\frac{1}{20}} - 1} = 0.2264312178$$

$$f\left(\frac{3}{20}\right) = \sqrt{e^{\frac{3}{20}} - 1} = 0.4022862696$$

$$f\left(\frac{5}{20}\right) = \sqrt{e^{\frac{5}{20}} - 1} = 0.53294035$$

$$f\left(\frac{7}{20}\right) = \sqrt{e^{\frac{7}{20}} - 1} = 0.6473542682$$

$$f\left(\frac{9}{20}\right) = \sqrt{e^{\frac{9}{20}} - 1} = 0.7538648324$$

$$f\left(\frac{11}{20}\right) = \sqrt{e^{\frac{11}{20}} - 1} = 0.8563019432$$

$$f\left(\frac{13}{20}\right) = \sqrt{e^{\frac{13}{20}} - 1} = 0.9568389776$$

$$f\left(\frac{15}{20}\right) = \sqrt{e^{\frac{15}{20}} - 1} = 1.056882215$$

$$f\left(\frac{17}{20}\right) = \sqrt{e^{\frac{17}{20}} - 1} = 1.157431143$$

$$f\left(\frac{19}{20}\right) = \sqrt{e^{\frac{19}{20}} - 1} = 1.259249641$$

Substitute the calculated values into the midpoint rule

$$M_{10} = \frac{1}{20} [0 + 0.2264312178 + 0.4022862696 + 0.53294035 + 0.6473542682 + 0.7538648324 + 0.8563019432 + 0.9500000000]$$

Therefore

$$M_{10} \approx 0.784958$$

c. Simpson's Rule to approximate the given integral with the specified value of  $n$ .

The coefficient pattern for  $n = 10$  is

$$1, 4, 2, 4, 2, 4, 2, 4, 2, 4, 1$$

So

$$S_{10} = \frac{\Delta x}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + 4f(x_5) + 2f(x_6) + 4f(x_7) + 2f(x_8) + 4f(x_9) + f(x_{10})]$$

Since

$$\frac{\Delta x}{3} = \frac{\frac{1}{10}}{3} = \frac{1}{30}$$

We have

$$S_{10} = \frac{1}{30} [0 + 4(0.32430066) + 2(0.4705345451) + 4(0.5914886369) + 2(0.7013021443) + 4(0.8054323502) + 2(0.9000000000)]$$

Therefore

$$S_4 \approx 0.780895$$

## Question 22

How large should  $n$  be to guarantee that the Simpson's Rule approximation to  $\int_0^1 e^{x^2} dx$  is accurate to within 0.00001?

Find the fourth derivative

$$f(x) = e^{x^2}$$

$$f'(x) = 2xe^{x^2}$$

$$f''(x) = e^{x^2}(4x^2 + 2)$$

$$f'''(x) = e^{x^2}(8x^3 + 12x)$$

$$f^{(4)}(x) = e^{x^2}(16x^4 + 48x^2 + 12)$$

Find  $K$  on the interval  $0 \leq x \leq 1$

all the terms in  $f^{(4)}(x)$  increases, so maximum occurs at  $x = 1$

Therefore

$$K = f^{(4)}(1)$$

$$K = 76e, \quad a = 0, \quad b = 1, \quad b - a = 1$$

$$|E_s| \leq 0.00001$$

Substitute into the error bound

$$\frac{76e(1)^5}{180n^4} \leq 0.00001$$

Simplify

$$\frac{76e}{180n^4} \leq 0.00001$$

Multiply by  $180n^4$

$$76e \leq 0.0018n^4$$

Divide by 0.0018

$$n^4 \geq 114771.8994$$

Take the fourth root

$$n \geq \sqrt[4]{114771.8994} = 18.40597773$$

Simpson's Rule requires  $n$  to be even.

The smallest even integer satisfying  $n \geq 18.40597773$

$$\boxed{n = 20}$$

So 20 subintervals guarantee that the Simpson's Rule error is at most 0.00001.

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### Question 30

The widths (in meters) of a kidney-shaped swimming pool were measured at 2-meter intervals as indicated in the figure. Use Simpson's Rule with  $n = 8$  to estimate the area of the pool.

Start with

$$f(x_0) = 0; \quad \Delta x = 2$$

|                  |   |     |     |     |     |     |     |     |    |
|------------------|---|-----|-----|-----|-----|-----|-----|-----|----|
| $x$ (m)          | 0 | 2   | 4   | 6   | 8   | 10  | 12  | 14  | 16 |
| Width $f(x)$ (m) | 0 | 6.2 | 7.2 | 6.8 | 5.6 | 5.0 | 4.8 | 4.8 | 0  |

Simpson's Rule

$$S_8 = \frac{2}{3} \left[ 0 + 4(6.2) + 2(7.2) + 4(6.8) + 2(5.6) + 4(5.0) + 2(4.8) + 4(4.8) + 0 \right]$$

$$\boxed{S_8 = 84.266667m^2}$$


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### Group 7.8

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### Question 5

Determine whether the integral is convergent or divergent. Evaluate integrals that are convergent.

$$\int_1^{\infty} 2x^{-3} dx$$

Rewrite as a limit

$$\int_1^{\infty} 2x^{-3} dx = \lim_{t \rightarrow \infty} \int_1^t 2x^{-3} dx$$

Find the antiderivative

$$\int 2x^{-3} dx = -x^{-2}$$

$$\lim_{t \rightarrow \infty} [-x^{-2}]_1^t$$

Evaluate the bounds

$$\lim_{t \rightarrow \infty} \left( -\frac{1}{t^2} - 1 \right)$$

Since

$$\lim_{t \rightarrow \infty} \frac{1}{t^3} = 0$$

$$-0 + 1 = 1$$

The limit exists and is finite

The improper integral is convergent.

$$\int_1^{\infty} 2x^{-3} dx = 1$$

### Question 6

Determine whether the integral is convergent or divergent. Evaluate integrals that are convergent.

$$\int_{-\infty}^{-1} \frac{1}{\sqrt[3]{x}} dx$$

Rewrite as a limit

$$\int_{-\infty}^{-1} \frac{1}{\sqrt[3]{x}} dx = \lim_{t \rightarrow -\infty} \int_{-\infty}^{-1} \frac{1}{\sqrt[3]{x}} dx$$

Find the antiderivative

$$\int \frac{1}{\sqrt[3]{x}} dx = \frac{3}{2} x^{\frac{2}{3}}$$

$$\lim_{t \rightarrow -\infty} \left[ \frac{3}{2} x^{\frac{2}{3}} \right]_t^{-1}$$

Evaluate the bounds

$$\lim_{t \rightarrow -\infty} \left( \frac{3}{2} t^{\frac{2}{3}} - \frac{3}{2} \right)$$

$$\lim_{t \rightarrow -\infty} \frac{3}{2} t^{\frac{2}{3}} = -\infty$$

The limit is not finite therefore

The improper integral is divergent.

---

### Question 8

Determine whether the integral is convergent or divergent. Evaluate integrals that are convergent.

$$\int_1^{\infty} \left( \frac{1}{3} \right)^x dx$$

Rewrite as a limit

$$\int_1^{\infty} \left( \frac{1}{3} \right)^x dx = \lim_{t \rightarrow \infty} \int_1^t \left( \frac{1}{3} \right)^x$$

Find the antiderivative

$$\int \left( \frac{1}{3} \right)^x = \frac{\left( \frac{1}{3} \right)^x}{\ln \left( \frac{1}{3} \right)}$$

$$\lim_{t \rightarrow \infty} \left[ \frac{\left( \frac{1}{3} \right)^x}{\ln \left( \frac{1}{3} \right)} \right]$$

Evaluate the bounds

$$\lim_{t \rightarrow \infty} \left[ \frac{\left( \frac{1}{3} \right)^t}{\ln \left( \frac{1}{3} \right)} - \frac{\frac{1}{3}}{\ln \left( \frac{1}{3} \right)} \right]_1^t$$

Since

$$\lim_{t \rightarrow \infty} \left( \frac{1}{3} \right)^t = 0$$

The limit is finite therefore

The improper integral is convergent.

$$\int_1^{\infty} \left(\frac{1}{3}\right)^x dx = \frac{1}{3 \ln 3}$$

---

### Question 9

Determine whether the integral is convergent or divergent. Evaluate integrals that are convergent.

$$\int_{-2}^{\infty} \frac{1}{x+4} dx$$

Rewrite as a limit

$$\int_{-2}^{\infty} \frac{1}{x+4} dx = \lim_{t \rightarrow \infty} \int_{-2}^t \frac{1}{x+4} dx$$

Find the antiderivative

$$\int \frac{1}{x+4} dx = \ln |x+4|$$

$$\lim_{t \rightarrow \infty} [\ln |x+4|]_{-2}^t$$

Evaluate the bounds

$$\ln |t+4| - \ln 2$$

As  $t \rightarrow \infty$

$$\ln |t+4| \rightarrow \infty$$

$$\int_{-2}^{\infty} \frac{1}{x+4} dx = \ln |x+4| = \infty$$

The limit is not finite therefore

The improper integral is divergent.

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## Question 12

Determine whether the integral is convergent or divergent. Evaluate integrals that are convergent.

$$\int_0^{\infty} \frac{1}{\sqrt[4]{1+x}} dx$$

Rewrite as a limit

$$\int_0^{\infty} \frac{1}{\sqrt[4]{1+x}} dx = \lim_{t \rightarrow \infty} \int_0^t \frac{1}{\sqrt[4]{1+x}} dx$$

Rewrite using a fractional exponent

$$\lim_{t \rightarrow \infty} \int_0^t (1+x)^{-\frac{1}{4}} dx$$

Find the antiderivative

$$\lim_{t \rightarrow \infty} \left[ \frac{4}{3} (1+x)^{\frac{3}{4}} \right]_0^t$$

Evaluate the bounds

$$\left[ \frac{4}{3} (1+t)^{\frac{3}{4}} - \frac{4}{3} (2)^{\frac{3}{4}} \right]$$

As  $t \rightarrow \infty$

$$\frac{4}{3} (1+t)^{\frac{3}{4}} \rightarrow \infty$$

Therefore

$$\lim_{t \rightarrow \infty} \left[ \frac{4}{3} (1+t)^{\frac{3}{4}} - \frac{4}{3} (2)^{\frac{3}{4}} \right] = \infty$$

Since the limit is not finite

The improper integral is divergent

---

### Question 16

Determine whether the integral is convergent or divergent. Evaluate integrals that are convergent.

$$\int_2^{\infty} \frac{x}{\sqrt{x^2 - 1}} dx$$

Rewrite as a limit

$$\int_2^{\infty} \frac{x}{\sqrt{x^2 - 1}} dx = \lim_{t \rightarrow \infty} \int_2^t \frac{x}{\sqrt{x^2 - 1}} dx$$

Apply  $u$ -substitution

Let

$$u = x^2 - 1 \quad du = 2x dx \quad \frac{1}{2} du = x dx$$

Rewrite the integral in terms of  $u$

$$\frac{1}{2} \int u^{-\frac{1}{2}} du$$

Find the antiderivative

$$\frac{1}{2} \left( \frac{u^{\frac{1}{2}}}{\frac{1}{2}} \right) = u^{\frac{1}{2}}$$

Substitute back  $u = x^2 - 1$

$$\sqrt{x^2 - 1}$$

Evaluate at the bounds

$$\lim_{t \rightarrow \infty} \left[ \sqrt{x^2 - 1} \right]_2^t$$

$$\lim_{t \rightarrow \infty} \left[ \sqrt{t^2 - 1} - \sqrt{3} \right]$$

As  $t \rightarrow \infty$

$$\sqrt{t^2 - 1} \rightarrow \infty$$

Therefore

$$\lim_{t \rightarrow \infty} [\sqrt{t^2 - 1} - \sqrt{3}] = \infty$$

The limit is not finite, therefore

The improper integral is divergent.

### Question 21

Determine whether the integral is convergent or divergent. Evaluate integrals that are convergent.

$$\int_{-\infty}^{\infty} \cos(2t) dt$$

Split the integral at 0

$$\int_{-\infty}^{\infty} \cos(2t) dt = \int_{-\infty}^0 \cos(2t) dt + \int_0^{\infty} \cos(2t) dt$$

Both improper integrals must converge independently for the original integral to converge.

Evaluate the left integral Rewrite as a limit

$$\int_{-\infty}^0 \cos(2t) dt = \lim_{a \rightarrow -\infty} \int_a^0 \cos(2t) dt$$

Find the antiderivative

$$\lim_{a \rightarrow -\infty} \int_a^0 \cos(2t) dt = \frac{1}{2} \sin(2t)$$

Evaluate the bounds

$$\lim_{a \rightarrow -\infty} \left[ \frac{1}{2} \sin(2t) \right]_a^0$$

$$\lim_{a \rightarrow -\infty} \left[ 0 - \frac{1}{2} \sin(2a) \right]$$

$$\lim_{a \rightarrow -\infty} \left[ -\frac{1}{2} \sin(2a) \right]$$

As  $a \rightarrow -\infty$ ,  $\sin(2a)$  continues to oscillate between  $-1$  and  $1$ . Therefore

$$\lim_{a \rightarrow -\infty} \left[ -\frac{1}{2} \sin(2a) \right] = DNE$$

Thus the left improper integral diverges. Since both halves must converge, there is no need to evaluate the right-hand integral.

The improper integral is divergent.

### Question 27

Determine whether the integral is convergent or divergent. Evaluate integrals that are convergent.

$$\int_{-\infty}^0 z e^{2z} dz$$

Rewrite as a limit

$$\int_{-\infty}^0 z e^{2z} dz = \lim_{t \rightarrow -\infty} \int_t^0 z e^{2z} dz$$

Apply integration by parts

$$\int u dv = uv - \int v du$$

let

$$u = z \quad dv = e^{2z}$$

$$du = dz \quad v = \frac{1}{2} e^{2z} dz$$

Therefore

$$\int z e^{2z} dz = \frac{z}{2} e^{2z} - \int \frac{1}{2} e^{2z} dz$$

Evaluate at the bounds

$$\lim_{t \rightarrow -\infty} \left[ \frac{z}{2} e^{2z} - \frac{1}{4} e^{2z} \right]_t^0$$

$$\lim_{t \rightarrow -\infty} \left[ -\frac{1}{4} - \left( \frac{t}{2} e^{2t} - \frac{1}{4} e^{2t} \right) \right]$$

As  $t \rightarrow -\infty$

$$e^{2t} \rightarrow 0$$

and

$$te^{2t} \rightarrow 0$$

Therefore

$$\lim_{t \rightarrow -\infty} \left[ -\frac{1}{4} - \left( \frac{t}{2}e^{2t} - \frac{1}{4}e^{2t} \right) \right] = -\frac{1}{4}$$

Since the limit is finite

The improper integral is convergent.

$$\int_{-\infty}^0 ze^{2z} dz = -\frac{1}{4}$$

---

### Question 29

Determine whether the integral is convergent or divergent. Evaluate integrals that are convergent.

$$\int_1^{\infty} \frac{\ln(x)}{x} dx$$

Rewrite as a limit

$$\int_1^{\infty} \frac{\ln(x)}{x} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{\ln(x)}{x} dx$$

Apply  $u$ -substitution Let

$$u = \ln(x) \quad du = \frac{1}{x} dx$$

Rewrite the integral

$$\int u du$$

Find the antiderivative

$$\frac{u^2}{2}$$

Substitute back  $u = \ln(x)$

$$\frac{(\ln x)^2}{2}$$

Evaluate at the bounds

$$\lim_{t \rightarrow \infty} \left[ \frac{(\ln x)^2}{2} \right]_1^t$$

$$\lim_{t \rightarrow \infty} \left[ \frac{(\ln t)^2}{2} - \frac{(\ln 1)^2}{2} \right]$$

$$= \lim_{t \rightarrow \infty} \left[ \frac{(\ln t)^2}{2} \right]$$

As  $t \rightarrow \infty$

$$\frac{(\ln t)^2}{2} \rightarrow \infty$$

Therefore

$$\lim_{t \rightarrow \infty} \int_1^t \frac{\ln(x)}{x} dx = \infty$$

Since the limit is not finite

The improper integral is divergent.

### Question 37

Determine whether the integral is convergent or divergent. Evaluate integrals that are convergent.

$$\int_{-2}^{14} \frac{dx}{\sqrt[4]{x+2}}$$

Identify the improper endpoint At  $x = -2$

$$\sqrt[4]{x+2} = \sqrt[4]{0} = 0$$

Therefore the integrand is undefined at that endpoint

Rewrite as a limit

$$\int_{-2}^{14} \frac{dx}{\sqrt[4]{x+2}} = \lim_{t \rightarrow -2^+} \int_t^{14} \frac{dx}{\sqrt[4]{x+2}}$$

Rewrite using a fractional exponent

$$\lim_{t \rightarrow -2^+} \int_t^{14} (x+2)^{-\frac{1}{4}} dx$$

Find the antiderivative

$$\lim_{t \rightarrow -2^+} \left[ \frac{4}{3} (x+2)^{\frac{3}{4}} \right]_t^{14}$$

Evaluate the bounds

$$\lim_{t \rightarrow -2^+} \left[ \frac{4}{3} (14+2)^{\frac{3}{4}} - \frac{4}{3} (t+2)^{\frac{3}{4}} \right]$$

$$\lim_{t \rightarrow -2^+} \left[ \frac{32}{3} - \frac{4}{3} (t+2)^{\frac{3}{4}} \right]$$

As  $t \rightarrow -2^+$

$$\frac{4}{3} (t+2)^{\frac{3}{4}} \rightarrow 0$$

Therefore

$$\lim_{t \rightarrow -2^+} \left[ \frac{32}{3} - \frac{4}{3} (t+2)^{\frac{3}{4}} \right] = \frac{32}{3}$$

Since the limit is finite

The improper integral is convergent.

Therefore

$$\boxed{\int_{-2}^{14} \frac{dx}{\sqrt[4]{x+2}} = \frac{32}{3}}$$

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### Question 42

Determine whether the integral is convergent or divergent. Evaluate integrals that are convergent.

$$\int_0^5 \frac{w}{w-2} dw$$

Identify the discontinuity The denominator equals zero when

$$w - 2 = 0$$

$$w = 2$$

Since  $w = 2$  lies inside the interval  $[0, 5]$ , split the integral

$$\int_0^5 \frac{w}{w-2} dw = \int_0^2 \frac{w}{w-2} dw + \int_2^5 \frac{w}{w-2} dw$$

Both improper integrals must converge for the original integral to converge.

Evaluate the left integral

Approach 2 from the left

$$\int_0^2 \frac{w}{w-2} dw = \lim_{t \rightarrow 2^-} \int_0^t \frac{w}{w-2} dw$$

Perform polynomial division

Since the numerator and denominator have the same degree, divide first

$$\frac{w}{w-2} = 1 + \frac{2}{w-2}$$

Find the antiderivative

$$\int \left( 1 + \frac{2}{w-2} \right) dw = w + 2 \ln |w-2|$$

Evaluate at the bounds

$$\lim_{t \rightarrow 2^-} [w + 2 \ln |w-2|]_0^t$$

$$\lim_{t \rightarrow 2^-} [t + 2 \ln |t-2| - 2 \ln 2]$$

As  $t \rightarrow 2^-$

$$|t - 2| \rightarrow 0^+$$

and

$$\ln |t - 2| \rightarrow -\infty$$

Therefore

$$\lim_{t \rightarrow 2^-} [t + 2 \ln |t - 2| - 2 \ln 2] = -\infty$$

Thus the left-hand improper integral diverges. Since both sides must converge there is no need to evaluate the right-hand integral.

The improper integral is divergent.

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